A Valid First-Law Probe at Fixed H: 1d Selects Enclosed Energy; 3d Balls Select CHM

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Abstract

Paper 36 showed that a Hamiltonian point source is the wrong variation: $\delta S \neq \mathrm{Tr}(K_{\mathrm{vac}}\Delta C)$. This note never changes H. Occupation α is moved from a localized occupied packet to its bipartite stagger.

The instrument passes. In 1d ($N = 200$, $L = 16$), $\rho(\delta S, \mathrm{Tr}(K_{\mathrm{vac}}\Delta C)) = 0.957$, but P_{flat} beats CHM ($R = 0.060$ vs 0.266; $\rho_{\mathrm{flat}} = 0.998$). Auditor: $\mathrm{C}_{\mathrm{vac}}$ CONFIRMED, C2 REFUTED.

On 512 balls of radius 2 ($N = 12$), the same construction selects CHM: $R = 0.018$ vs flat 0.102 vs linear 0.029. $\mathrm{C}_{\mathrm{vac}} = 0.999$. Auditor $N = 10$: C2 CONFIRMED.

A linear functional, yes. Einstein's equation, no. The 1d theorem case still prefers Gauss, not the boost weight. Not $8\pi G$. Not de Sitter.

1 What this is

K itself can be CHM-shaped (Paper 25) while the pairing of this excitation against K is the characteristic function of a 1d interval and the CHM weight on a small 3d ball. That is the first-law half of Faulkner–Guica–Hartman–Myers–Van Raamsdonk as a lattice measurement of one variation. It is not a derivation of $G_{\mu\nu} = 8\pi G T_{\mu\nu}$.

Admission. *I repaired the variation until the vacuum first law held. I did not hide that 1d still selects enclosed energy, and I did not write Einstein.*